

Discrete-Continuous Dynamic Choice Models:

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Contribution

Framework: Simultaneous discrete-continuous choice

- Core identification problem: only observe continuous choice for the chosen discrete alternative

Identification: Constructive proof of nonparametric identification

- Instrument: *relevant* for discrete choice, *excluded* from continuous choice

Estimation: Two-step estimation

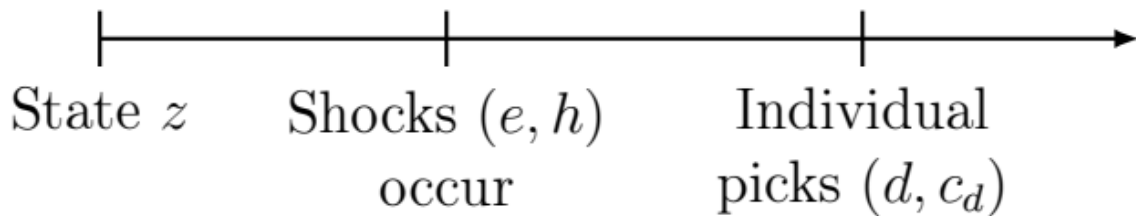
- 1) read CCC and CCP from data
- 2) structural estimation to recover parameters

Plan

- **Framework:** Static
- **Identification:** Static
- **Framework:** Dynamic
- **Identification:** Dynamic

Framework: Static

Timeline



Shocks:

- $e = (e_0, e_1) \in \mathbb{R}^2$ only discrete choice for one alternative, drawn from ϵ
- $h \in \mathbb{R}$ shocks discrete and continuous for both alternatives, drawn from η

State Variables: z

Choices:

- $d \in 0, 1$ discrete choice
- $c_d \in \mathbb{R}$ continuous choice given d

Payoffs:

$$\max_{d, c_d} V_d(c_d, z, h, e_d)$$

Framework: Assumptions

1. **Additive Separability:** e enters the payoff additively, for each d

$$V_d(c_d, z, h, e_d) = \tilde{v}_d(c_d, z, h) + e_d$$

Assumption 1 implies: conditional policy c_d^* does not depend on e

2. **Instrument:** $z = (x, w)$ and w does not enter the conditional continuous choice

$$\tilde{v}_d(c_d, z, h) = v_d(c_d, x, h) + m_d(x, w, h)$$

Assumption 2 implies: conditional policy c_d^* does not depend on w

Framework: More Assumptions

3. **Monotonicity:** v_d is twice continuously differentiable and $\frac{\partial^2 v_d(c_d, z, h)}{\partial c_d \partial h} > 0$

Assumption 3 implies:

- given d and x , $c_d^*(z, h)$ is one-to-one with h
 - the framework requires non-trivial continuous choice for each discrete choice
4. **Shocks:** Conditional on $X = x$, (i) W, η, ϵ are mutually independent, (ii) η drawn from $Uniform(0, 1)$, (iii) ϵ is continuously distributed with full support/, and (iv) $\max_c \tilde{v}_d(c, x, w, h) < \infty$ for all (x, w, h, d) .
5. **Instrument Relevance:** For every x , $P(D = 0 \mid \eta = h, X = x, W = 1) \neq P(D = 0 \mid \eta = h, X = x, W = 0)$, for all $h \in \mathcal{H} \setminus \mathcal{K}_x$, where $\mathcal{K}_x$ is a possibly empty finite set.

Framework: Example - Static Energy Demand

Decision Problem:

$$\max_{d, c_d} \{v_d(c_d, x, h) + m_d(x, w, h) + e_d\}$$

Energy Sources: Discrete $d = 0$ or 1

Energy Demand: Continuous c_0 and c_1

Characteristics: x , cost, income, etc

Instrument: w , e.g. previous energy source

Taste Shock: e_d specific to sources

Demand Shock: h , shifts c_d for $d = 0$ and 1 , also enters the discrete choice

Framework: Example - Implication of Assumptions

Decision Problem:

$$\max_{d, c_d} \left\{ v_d(c_d, x, h) + \underbrace{m_d(x, w, h) + e_d}_{\text{discrete choice shock}} \right\}$$

Instrument Shifts Conditional Choice Probability:

$$\begin{aligned} & P(D = 0 \mid \eta = h, X = x, W = 1) \\ &= P(e_0 - e_1 > \max_{c_1} v_1(c_1, x, h) - \max_{c_0} v_0(c_0, x, h) + \\ & \quad m_1(x, w, h) - m_0(x, w, h) \mid \eta = h, X = x, W = w) \end{aligned}$$

Framework: Example - Implication of Assumptions

Instrument Shifts Conditional Choice Probability:

$$\begin{aligned} & P(D = 0 \mid \eta = h, X = x, W = 1) \\ &= P(e_0 - e_1 > \max_{c_1} v_1(c_1, x, h) - \max_{c_0} v_0(c_0, x, h) + \\ & \quad m_1(x, w, h) - m_0(x, w, h) \mid \eta = h, X = x, W = w) \end{aligned}$$

since $(e_0, e_1) \perp (W, \eta) \mid X = x$ (A4) and

$\max_{c_1} v_1(c_1, x, h) - \max_{c_0} v_0(c_0, x, h)$ independent of w (A2).

We have that (A5) is equivalent to an assumption on discrete choice shock:

$$\begin{aligned} & P(D = 0 \mid \eta = h, X = x, W = 1) \neq P(D = 0 \mid \eta = h, X = x, W = 0) \\ & \Leftrightarrow m_1(x, w = 1, h) - m_0(x, w = 1, h) \neq m_1(x, w = 0, h) - m_0(x, w = 0, h) \end{aligned}$$

Identification: Static

Data: (D, C, W) , we omit X as it is exogenous and ϵ, W, η mutually independent when fixing on $X = x$

Structures to be Identified:

-Conditional Continuous Choices: $c_d^*(h)$

-Conditional Choice Probability: $P(D = d \mid \eta = h, W = w)$

-Indirect Payoffs: $\max_c v_d(c, h)$ and $m_d(w, h)$

Reduced Forms Recovered from Data:

- Distribution of Continuous Choice given d and w : $F_{C_d | d, w}(c_d) = P(C_d < c_d | d, w)$
- Choice Probability given w : $p_{D | w}(d) = P(d | w)$

Identification: Lemmas

Lemma 1: Under Assumption 3 and 4, $F_{C_d | d, w}(c_d) = P(C_d < c_d | d, w)$ is continuously differentiable and strictly increasing, for $d = 0, 1$.

-Intuition: Monotonicity implies that h and $c_d^*(h)$ are one-to-one given d, w . We have that $F_{\eta | d, w}$ is continuously differentiable and strictly increasing, so $F_{C_d | d, w}$ inherits these properties.

Lemma 2: Under Assumption 5, $\frac{\partial F_{C_d | d, 1}(c_d) p_{D | 1}(d)}{\partial c_d} \neq \frac{\partial F_{C_d | d, 0}(c_d) p_{D | 0}(d)}{\partial c_d}$ for $d = 0, 1$, where $h \in \mathcal{H} \setminus \mathcal{K}_x$, where $\mathcal{K}_x$ is a possibly empty finite set.

-Intuition: Given the same discrete choice, individual with different w experienced different distributions of h shock. Since h and $c_d^*(h)$ are one-to-one, we can take this restriction to data.

Identification: Conditional Continuous Choice

Goal: Identify $c_0^*(h)$ and $c_1^*(h)$

Challenge: $P(\eta < h | D=d) = P(C_d < c_d^*(h) | D=d) \implies c_d^*(h) = F_{C_d | d}^{-1}(P(\eta < h | D=d))$ unobserved!

Identification Intuition: Fixing an h and $w = 1$, we have

$$\begin{aligned} h &= P(\eta < h) \\ &= P(\eta < h | W = 1) \quad [A4] \\ &= P(\eta < h | D = 0, W = 1)P(D = 0 | W = 1) + \\ &\quad P(\eta < h | D = 1, W = 1)P(D = 1 | W = 1) \\ &= P(C_0 < c_0^*(h) | D = 0, W = 1)P(D = 0 | W = 1) + \\ &\quad P(C_1 < c_1^*(h) | D = 1, W = 1)P(D = 1 | W = 1) \\ &= F_{C_0 | 0, 1}(c_0^*(h))p_{D | 1}(0) + F_{C_1 | 1, 1}(c_1^*(h))p_{D | 1}(1) \end{aligned}$$

Identification: Conditional Continuous Choice

Identification Intuition: Fixing an h and $w = 1$, we have

$$h = F_{C_0 | 0, 1}(c_0^*(h))p_{D | 1}(0) + F_{C_1 | 1, 1}(c_1^*(h))p_{D | 1}(1)$$

Note that $F_{C_d | d, w}(\cdot)$ and $p_{D | w}(\cdot)$ are observed from reduced form. We have two unknowns: $c_0^*(h)$ and $c_1^*(h)$, but also an extra equation from $w = 0$. Thus we can solve for the CCCs for each h .

Theorem 1 For every reduced form compatible with the structural model, there exist unique conditional continuous choice (CCC) functions $c_d(h)$, $d = 1, 0$ that are strictly increasing and satisfy:

$$h = F_{C_0 | 0, w}(c_0(h))p_{D | w}(0) + F_{C_1 | 1, w}(c_1(h))p_{D | w}(1) \quad \text{for } h \in [0, 1], w = 0, 1$$

Intuition as above, proof establishes uniqueness.

Identification: Conditional Choice Probability and Payoffs

CCP: Identify $P(D = d | \eta = h, W = w)$

Challenge: η unobserved

Solution: Given $c_0^*(h)$ and $c_1^*(h)$ are identified and strictly monotonic, we can recover $\eta = (c_D^*(h))^{-1}(C)$ from data (D, C) . It is then as if (D, C, W, η) are all observed.

Payoffs: Identified, but depend on how the CCCs and CCPs relate to the utility/value function.

Framework: Dynamic

Assumptions D1-D3: Assumptions on the current utility, similar to the static version but with t subscript.

6. **Conditional Independence:** For all $x_t \in X, w_t \in W, e_t \in \epsilon, h_t \in H$,

$$f_{X_{t+1}, W_{t+1}, \epsilon_{t+1}, \eta_{t+1} | x_t, w_t, c_t, d_t, e_t, h_t}(x_{t+1}, w_{t+1}, e_{t+1}, h_{t+1}) = f_{X_{t+1}, W_{t+1} | x_t, w_t, c_t, d_t}(x_{t+1}, w_{t+1}) f_{\epsilon}(e_{t+1}) f_{\eta}(h_{t+1})$$

7. **Instrument Transition Exclusion:** For all $x_t \in X, w_t \in W$, the current instrument is excluded from the transition density:

$$f_{X_{t+1}, W_{t+1} | x_t, w_t, c_t, d_t}(x_{t+1}, w_{t+1}) = f_{X_{t+1}, W_{t+1} | x_t, c_t, d_t}(x_{t+1}, w_{t+1})$$

Assumption 6 and 7 are required to fit the dynamic problem into the general framework we developed before.

Framework: Dynamic

The Dynamic Decision Problem:

$$V_t(z_t) := \mathbb{E} \left[\sum_{\tau=t}^T \max_{D, C_{D\tau}} u_{D\tau}(C_{\tau}, X_{\tau}, \eta_{\tau}) + m_{D\tau}(X_{\tau}, W_{\tau}, \eta_{\tau}) + \epsilon_{D\tau} \right]$$

We can write it recursively, using assumption 6 and 7

$$V_t(z_t) := \mathbb{E}_{\epsilon, \eta} \left[\max_{D, C_{D\tau}} [u_{dt}(c_{dt}, x_t, \eta_t) + m_{dt}(x_t, w_t, \eta_t) + \epsilon_{dt} + \beta \mathbb{E}_{Z_{t+1}} [V_{t+1}(Z_{t+1}) | X_t = x_t, C_t = c_{dt}, D_t = d_t]] \right]$$

With the instrument excluded from the state transition, the continuation value term does not depend on w .

Framework: Dynamic

Define the conditional value function:

$$v_{dt}(c_{dt}, x_t, \eta_t) = u_{dt}(c_{dt}, x_t, \eta_t) + \beta \mathbb{E}_{Z_{t+1}} [V_{t+1}(Z_{t+1}) \mid X_t = x_t, C_t = c_{dt}, D_t = d_t]$$

Reformulate the per-period problem, we have:

$$\max_{d_t, c_{dt}} \{v_{dt}(c_{dt}, x_t, h_t) + m_{dt}(x_t, w_t, h_t) + e_{dt}\}$$

which is of the same form as our static problem.

Lemma 3: Under Assumptions D1-D3, 6, and 7, Assumption 1, 2, 3 are satisfied for the conditional value function in the dynamic setup.

-Intuition: Assumption 1 (AS) follows from D1.

Assumption 2 (Instrument) follows from D2 + 7 (Instrument Transition Exclusion).

Assumption 3 (Monotonicity) follows from D3 + 6 (CI).

$$\frac{\partial^2 v_{dt}(c_{dt}, x_t, h_t)}{\partial c_{dt} \partial h_t} = \underbrace{\frac{\partial^2 u_{dt}(c_{dt}, x_t, h_t)}{\partial c_{dt} \partial h_t}}_{>0, \text{ by } D3} + \underbrace{\frac{\partial^2 \mathbb{E}_{Z_{t+1}} [V_{t+1}(Z_{t+1}) \mid X_t = x_t, C_t = c_{dt}, D_t = d_t]}{\partial c_{dt} \partial h_t}}_{=0, \text{ by } CI} > 0$$

Framework: Dynamic Example

We still need an instrument, the author proposes the previous discrete choice $W_t = D_{t-1}$.

Decision Problem:

$$\max_{d_t, c_{dt}} \{v_{dt}(c_{dt}, x_t, h_t) + m_{dt}(x_t, w_t, h_t) + e_{dt}\}$$

Energy Sources: Discrete $d_t = 0$ or 1

Energy Demand: Continuous c_{0t} and c_{1t}

Characteristics: x , cost, income, etc

Instrument: w , e.g. previous energy source

Taste Shock: e_{dt} specific to sources

Demand Shock: h_t , shifts c_{dt} for $d = 0$ and 1, also enters the discrete choice

Framework: Dynamic Example

The author argues:

- 1) conditional on current source of energy d_t and x_t , future w_{t+1} is known and x_{t+1} is unlikely to be influenced by past choice [defending Assumption 7, Transition Exclusion]
- 2) conditional on current source of energy d_t and x_t , past choice unlikely to enter current utility function [defending Assumption D2, Exclusion]
- 3) past choice likely relevant if there is a switching cost [defending Assumption 5, Relevance]

Identification: Dynamic

As before, we are interested in identifying the CCCs c_{dt}^*

CCPs $P(D_t = d_t \mid \eta_t = h_t, X_t = x_t, W_t = w_t)$, the transition laws, and the value function parameters.

Data: (D_t, C_t, X_t, W_t, t)

Reduced Forms: $R = \left\{ P(D = d_t \mid X_t = x_t, W_t = w_t), F_{C_{dt} \mid d_t, x_t, w_t}(\cdot) \right\}$

For CCCs and CCPs, we proceed as before, exploiting our instrumental variable.

For transition laws are non-parametrically identified from data.

The value functions are identified by extra assumptions on the ϵ_t term and on the current utility function, using existing results.

More in the paper

-With unobserved types

-Two-step Estimation